

DUBOIS, PA, USA

WMO: 725125

Lat: 41.179N

Lon: 78.893W

Elev: 1805

StdP: 13.76

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 04787

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.6	5.6	-7.8	3.9	3.1	-2.9	5.0	7.6	26.3	28.4	24.3	25.3	9.4	260	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	84.7	70.0	82.3	68.8	80.5	67.6	72.9	80.9	71.3	78.6	70.0	76.7	9.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.3	119.4	76.9	69.1	114.2	75.4	67.7	108.7	73.9	37.6	81.1	36.2	78.7	35.1	76.9	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.3	19.5	17.6	-4.9	88.8	6.9	4.0	-9.8	91.6	-13.9	94.0	-17.7	96.2	-22.7	99.1
			-5.4	75.7	6.2	1.4	-9.8	76.7	-13.4	77.5	-16.9	78.3	-21.4	79.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.0	24.5	26.8	35.2	47.4	57.5	65.2	69.1	67.7	61.4	50.4	39.3	30.0
	DBStd	18.11	12.10	11.14	11.28	9.76	8.50	6.23	5.06	5.02	6.97	8.65	9.63	9.87
	HDD50	3192	795	652	479	162	27	1	0	0	4	104	343	625
	HDD65	6632	1257	1069	923	533	262	73	19	30	150	458	772	1086
	CDD50	2455	4	3	21	83	258	457	592	550	345	117	21	4
	CDD65	421	0	0	1	4	28	80	146	115	41	6	0	0
	CDH74	2793	0	0	3	41	237	520	982	707	276	27	0	0
	CDH80	535	0	0	0	3	33	89	216	138	53	3	0	0
Wind	WSAvg	8.2	10.4	9.8	9.5	9.3	7.8	6.7	6.0	5.7	6.3	7.9	8.9	9.7
Precipitation	PrecAvg	42.8	2.7	2.6	3.7	3.3	3.9	4.5	3.9	4.0	3.8	2.8	3.4	3.5
	PrecMax	56.5	6.5	5.5	6.7	4.8	9.1	14.7	9.4	7.0	5.9	5.2	8.7	7.0
	PrecMin	34.8	1.1	0.9	1.3	1.2	0.9	0.7	1.9	0.7	0.7	0.7	1.3	1.3
	PrecStd	4.4	1.3	1.2	1.4	1.1	1.9	3.0	1.9	1.6	1.4	1.2	1.6	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.7	61.4	71.6	79.1	83.7	85.6	88.4	87.2	85.3	77.4	67.7	60.9
		MCWB	53.5	50.7	56.4	60.6	68.0	70.6	71.8	70.2	69.7	64.8	54.9	55.0
	2%	DB	54.2	53.7	64.2	73.5	80.1	82.4	84.8	83.7	80.9	72.3	62.7	54.8
		MCWB	51.0	47.6	51.9	56.9	64.3	69.2	70.9	69.8	67.3	61.2	53.7	51.1
	5%	DB	47.7	47.6	58.1	68.4	76.5	80.1	82.1	81.0	76.9	68.5	58.4	49.4
		MCWB	44.4	41.5	48.8	54.1	62.5	67.7	69.4	68.2	65.5	59.7	51.2	45.2
	10%	DB	41.3	42.9	53.1	63.7	72.4	76.8	79.7	78.3	73.2	64.3	54.6	44.0
		MCWB	38.3	38.0	46.1	51.4	60.9	65.7	68.0	66.9	63.9	57.2	49.1	40.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.1	54.0	58.2	63.3	70.1	73.1	75.5	74.3	72.7	68.3	59.5	56.3
		MCDB	58.1	59.4	68.1	74.0	79.6	81.6	84.2	82.2	80.9	74.5	65.7	59.1
	2%	WB	51.2	48.0	54.6	59.6	67.5	71.2	73.4	72.2	69.7	64.6	55.6	51.4
		MCDB	54.1	52.9	61.0	68.6	76.4	79.4	81.7	79.3	76.5	69.0	60.1	53.6
	5%	WB	44.7	42.3	50.9	57.0	65.3	69.7	71.5	70.6	67.8	61.3	52.8	46.0
		MCDB	47.0	46.4	57.4	65.3	73.0	77.2	78.7	77.0	74.1	66.4	57.3	48.5
	10%	WB	38.1	37.9	45.9	54.0	63.0	68.0	70.1	69.0	65.8	58.2	49.4	40.6
		MCDB	40.9	42.2	52.5	61.5	70.1	74.5	76.6	75.0	71.2	63.5	54.7	43.8
Mean Daily Temperature Range	5% DB	MDBR	13.2	14.7	16.7	20.0	19.6	18.4	18.7	18.5	18.4	16.9	14.2	11.8
		MCDBR	17.2	20.7	23.7	25.7	23.2	21.3	20.8	21.2	22.0	21.5	19.9	18.5
		MCWBR	14.8	16.2	14.9	14.3	11.5	10.7	9.5	9.7	10.5	12.1	13.8	15.7
	5% WB	MCDBR	16.8	19.3	21.2	21.2	19.4	18.1	17.8	17.2	17.7	18.0	17.6	17.7
		MCWBR	15.5	16.5	15.4	13.9	10.7	10.1	9.3	9.0	9.5	11.7	14.3	16.5
Clear-Sky Solar Irradiance	taub	0.300	0.322	0.342	0.363	0.398	0.437	0.441	0.431	0.366	0.355	0.315	0.295	
	taud	2.410	2.326	2.370	2.367	2.281	2.193	2.196	2.217	2.383	2.413	2.506	2.484	
	Ebn at Noon	271	280	288	288	279	267	265	264	276	265	265	265	
	Edh at Noon	25	32	35	37	42	46	45	43	34	30	23	22	
All-Sky Solar Radiation	RadAvg	468	723	1039	1407	1617	1768	1799	1612	1305	856	568	373	
	RadStd	39	92	108	145	159	151	132	96	148	91	75	41	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+218	N/A
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+184	N/A

Nomenclature: See separate page